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Text Classification

Text Mining

Departemen Teknik Informatika

Informatics Department

Overview

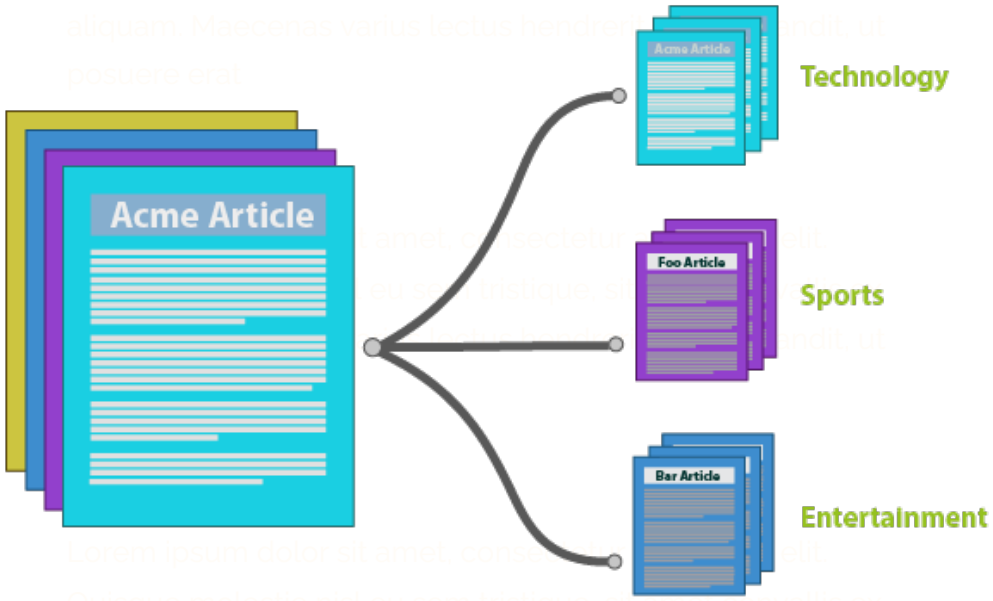


What is Text Classification?

Text classification is the task of assigning **predefined categories** or labels to text documents based on their content.

- It is a fundamental task in Natural Language Processing (NLP) that enables machines to interpret and organize textual information.

Goal: Convert unstructured text into structured, actionable information.



What is it important?

- Handles large-scale text data efficiently.
- Enables intelligent systems (e.g., chatbots, recommender systems).
- Fundamental in natural language processing (NLP).


Performance
Classification
Improve
Text
Task
Use
Model
Data
Accuracy
Multiple
Each
Specific
Ensemble
Different
Granular
Topic
Learning
Trained
More
Lead
Entity
Approach
Handling
Adaptation
Robust
Machine
System
Identify
Emotion
Becomes
Overall
Important
Interpretability
Noisy
Outliers
Negative
Combining
Words
Given
Domain
Pipeline
Result
Overfitting
Risk
Accurate
Help
Multi
Plot
Individual
Parallel
Discusses
Case
Describe
Advantage
Boosts
Classifies
Scalable
Natural
Visual
Miss
Handling
Strengths
Particular
Potential
Many
General
Focuses
Patterns
Gain
Rapid
Areas
Impact
Out
Errors
Nuances
Given
Particular
Suitable
Behind
Picture
Confidence
Suited
Labels
Each
Changes
Deployment
Support
Contributions
Efficient
Information
Prototyping
Help
Learning
Assessment
Good
Characters
System
Deep
Biases
Outputs
Descriptive
Occurs
Assign
Enhanced
Result
Focus
Topic
Specific
Manage
Prediction
Related
Time
Industry
Criterion
Resources
Ensemble
Drift
Large
Text
Unseen
Recognition
Quickly
Content
Wide
Highly
Less
Finance
Data
Solution
Strategies
Outliers
Associated
Granular
Improve
Aspect
Real
Pipeline
Benefits
Result
Enhanced
Efficient
Allows
Mitigated
Presence
Approach
Sensitive
Understanding
Challenges
Fewer
Saving
Tailor
Relevant
Way
Emotion
Faster
Accurate
Out
Leveraging
Depending
Entity
Analysis
Practitioners
Contains
Reviews
Combining
Given
Lead
Negative
Complete
Underlying
Main
Robust
Techniques
Stable
Outliers
Based
Provide
Comparing
Enable
Highlights
Reduced
Positive
Common
Detect
Maintain
Extract
Soundtrack
Requirements
Trees
Metrics
Various
Thresholds
Comes
Dataset
Movie
Decision
Instance
Deploying
Vector
Capable
Overall
Goal
World
Aspect
Article
Better
Trained
Model
Entity
Accurate
Range
Capture
Consider
Labels
Making
More
Effects
Distribution
Article
Needs
Minimize
Accuracy
Multiple
Experimentation
Jargon
Driving
Multiple
Domain
Particularly
Use
Recognition
Incremental
Offer
Adept
Customizable
Achieved
Determine
Additional
Insights
Computational
Classifiers
Words
Structure
Language
Maintain
Soundtrack
Extract
Requirements
Detect
Each
Various
Labels
Adaptation
Deploying
Decision
Instance
Interpretability
Noisy

Types of Text Classification

Binary Classification

- Spam vs Not Spam

Multiclass Classification

- Categorizing news into Sports, Politics, Tech, etc.

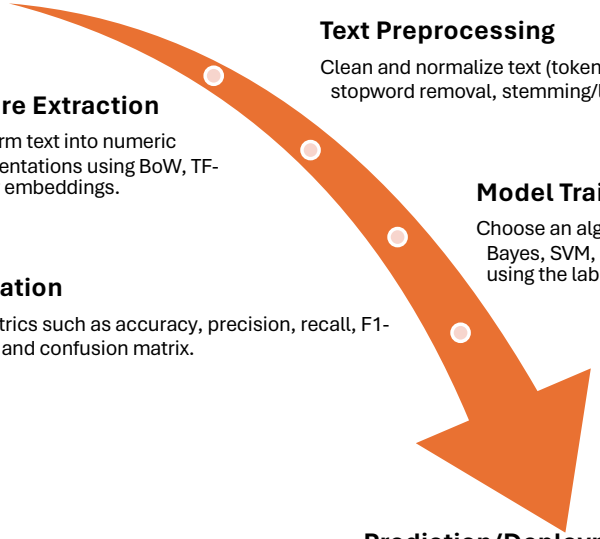
Multilabel Classification

- A document can belong to more than one class.

Text Classification Pipeline

Text Collection

Gather data from documents, websites, APIs, etc.



Text Preprocessing

Clean and normalize text (tokenization, lowercasing, stopword removal, stemming/lemmatization).

Feature Extraction

Transform text into numeric representations using BoW, TF-IDF, or embeddings.

Model Training

Choose an algorithm (e.g., Naive Bayes, SVM, BERT) and train it using the labeled data.

Evaluation

Use metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

Prediction/Deployment

Classify new, unseen text using the trained model.

Traditional Algorithms

Naive Bayes

- Probabilistic and simple.

Logistic Regression

- Linear model for binary and multiclass tasks.

Support Vector Machines (SVM)

- Effective in high-dimensional spaces.

K-Nearest Neighbors (KNN)

- Distance-based classifier.

Deep Learning Approaches

Recurrent Neural Networks (RNN, LSTM)

- Good for sequence data.

Convolutional Neural Networks (CNN)

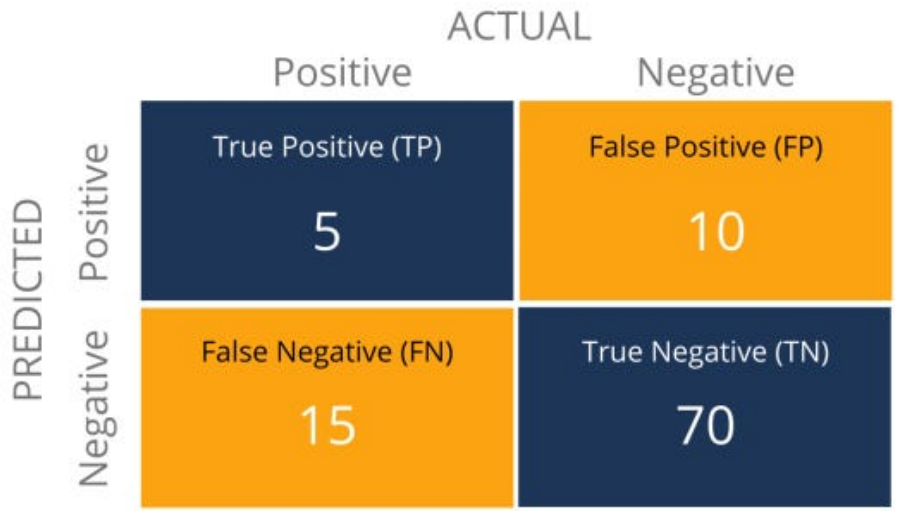
- Captures local patterns.

Transformers (BERT, RoBERTa)

- State-of-the-art performance.

Fine-tuning pre-trained language models.

Evaluation Metrics



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = 0.75$$

$$\text{Recall} = \frac{TP}{TP + FN} = 0.25$$

$$\text{Precision} = \frac{TP}{TP + FP} = 0.33$$

$$\text{F1 Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} = 0.28$$

Use case 1: Healthcare



Classify clinical notes into diagnostic categories

Identify mentions of symptoms, medications, or conditions

Use case 2: Legal



Categorize case files and legal documents

Extract legal clauses and classify contracts by type

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Use case 3: Marketing



- Analyze customer feedback and product reviews
- Segment audience based on behavior or preferences

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Challenges in Text Classification

- Class Imbalance
- Noisy Text (typos, slang, informal language)
- Ambiguity and Polysemy
- Domain-Specific Language
- Short Texts and Sparse Features

Other materials

Reading:

https://www.analyticsvidhya.com/blog/2018/04/a-comprehensive-guide-to-understand-and-implement-text-classification-in-python/

Code:

1. Multiclass classification with ML: https://www.kaggle.com/code/selener/multi-class-text-classification-tfidf
2. Binary classification with DL: https://www.kaggle.com/code/yacharki/binary-text-classification-yelp-92-15-cnn
3. Multilabel classification: https://www.kaggle.com/code/owaiskhan9654/multi-label-classification-of-pubmed-articles

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Thank you